

UMC 0.11um RFCMOS Al Advanced Enhancement Metal Oxide Metal Capacitors RF SPICE Model Document

**Version 0.2 Phase 2
2010/11/12**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2p1	2010/07/22	Bing Chang Wu	
0.2p2	2010/11/12	Bing Chang Wu	typo update

1.2 General Information

This document describes RF (Radio Frequency) MOM (Metal-Oxide-Metal) capacitor SPICE models, and this MOM is built on the metal 1~6 of UMC's L110AE process.

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

The capacitor models in this document are generated based on intensive EM (Electromagnetic) simulation. The EM simulation is well-calibrated based DSM data to ensure EM simulation accurately predicts capacitor performance.

The models should be used in conjunction with UMC capacitor Pcells (Programmable layout Cell) provided in UMC MM/RF FDK (Foundry Design Kit). The models may not be valid if the capacitor layout is different from UMC Pcell.

In current release, 3 major commercial simulators (Cadence's SPECTRE, Synopsys' HSPICE and Mentor Graphics' ELDO) are supported.

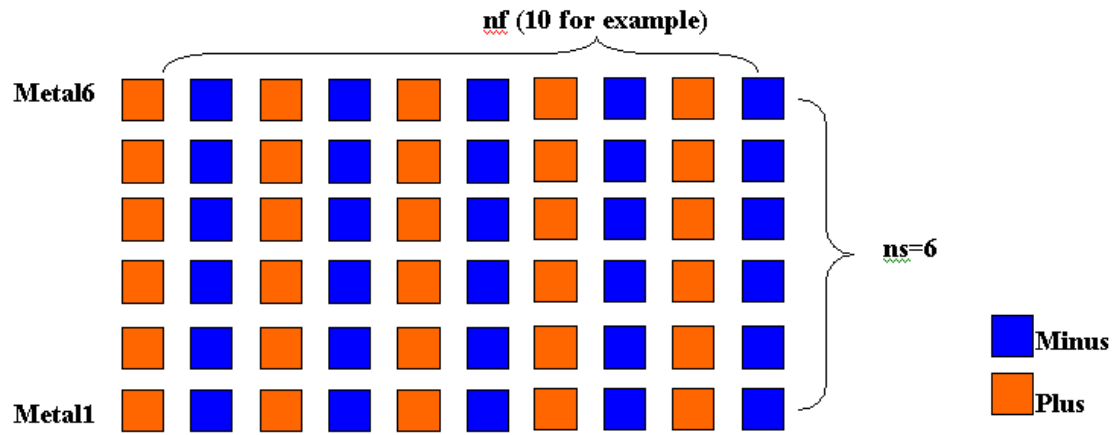
It is suggested to read the model document firstly to understand the model usage and limitation before using the models.

1.3 Device Structure

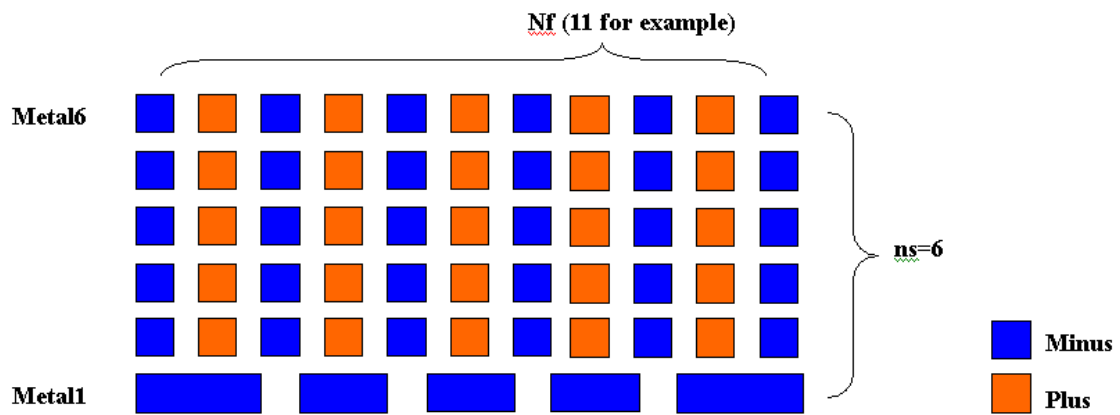
MOM capacitors rely on coupling capacitance between metal fingers running in parallel. Metal layers are stacked from bottom metal to upper metal layer to increase the capacitance density.

Metal finger width and spacing are fixed at the minimum dimensions, i.e. 0.2 μm (before shrinkage), to achieve the maximum capacitance density.

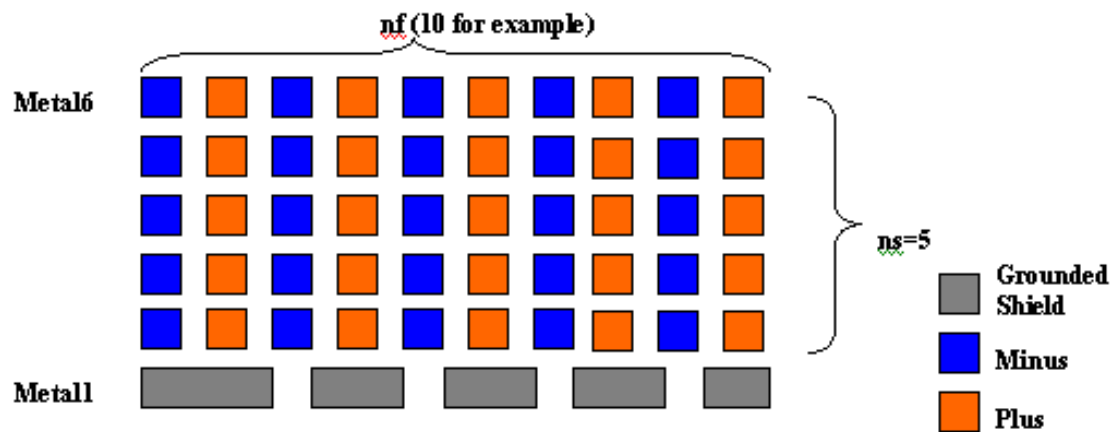
The architecture is basically symmetric in terms of the 2 ports (Plus and Minus) for VP3 and asymmetric in terms of the 2 ports (Plus and Minus) for VP4 and shielded in terms of the 2 ports (Plus and Minus) for VP5. The drawing layer of Plus port is (Bottom metal +1) and Minus is (Bottom metal +2). Only logic or mixed signal devices are allowed to put under VP5 MOM structure! And, there is no well type restriction in VP5 MOM structure.



(a) Example of VP3 Symmetric MOM



(b) Example of VP4 Asymmetric MOM



(c) Example of VP5 Shielded MOM

Figure 1-1 Sample cross section of RFMOM

1.4 Model Usage

It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2007.09

Cadence SPECTRE version 6.2.1.299

Mentor Graphics ELDO version 6.11_1.1

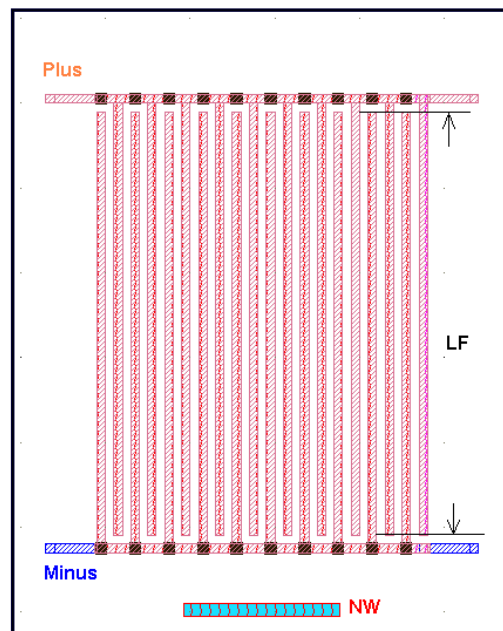
To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

ff : minimum C value

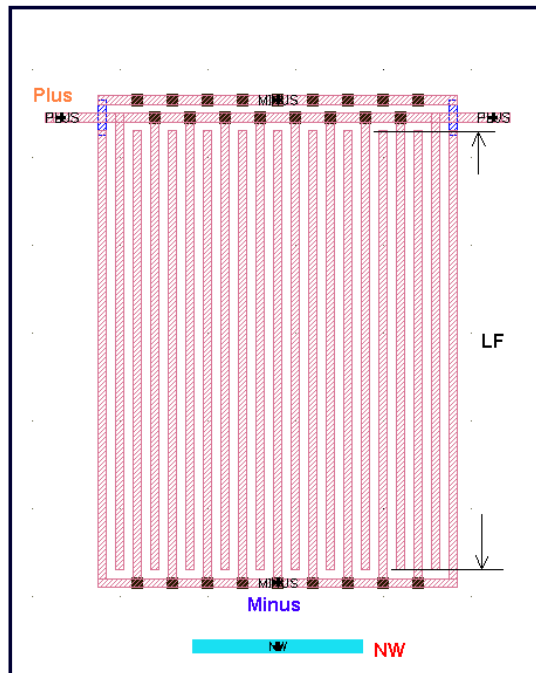
tt : typical C value

ss : maximum C value

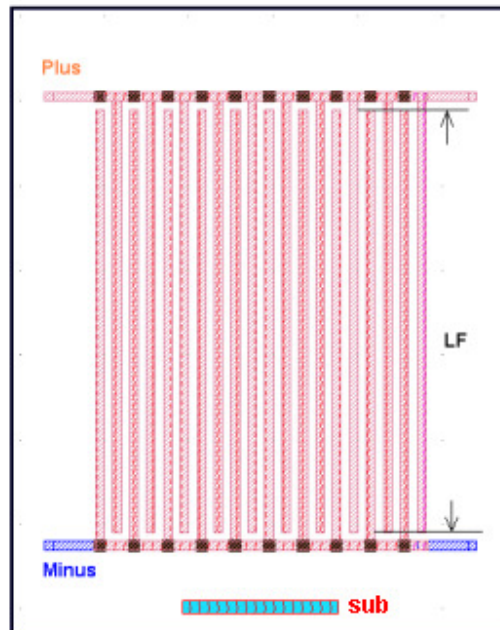
Follow the description for net-list of RFMOM, and these steps to invoke the model file during simulation.



(a) Example for Symmetric MOM (VP3, Even Finger Number, and here NF=20)



(b) Example for Asymmetric MOM (VP4, Odd Finger Number, and here NF=21)



(c) Example for Shielded MOM (VP5, Even Finger Number, and here NF=20)

Figure 1-2 Sample layout and net-list of RFMOM

For Hspice

- Copy ModelFileName.lib & ModelFileName.mdl into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "/ModelFileName.lib" CornerSection
, where CornerSection could be ff, tt, ss.
- Follow the example below to call the model

Example 1: For VP3 MOM (Refer to Figure 1-1 (a))

x1 plus minus nwell sub momcaps_array_vp3_rf nf=20 lf=20u bm=1 ns=4
array=1 m=5

Example 2: For VP4 MOM (Refer to Figure 1-1 (b))

x1 plus minus nwell sub momcaps_array_vp4_rf nf=21 lf=20u bm=1 ns=4
array=1 m=5

Example 3: For VP5 MOM (Refer to Figure 1-1 (c))

x1 plus minus sub momcaps_array_vp5_rfvcl nf=20 lf=20u bm=1 ns=4 array=1
m=5

For Cadence Spectre

- Copy ModelFileName.lib.scs & ModelFileName.mdl.scs into your simulation directory.
- Add the following statement in the input netlist to the simulator
include "/ModelFileName.lib.scs" section=CornerSection
, where CornerSection could be ff, tt, ss.
- Follow the example below to call the model

Example 1: For VP3 MOM (Refer to Figure 1-1 (a))

x1 (plus minus nwell sub) momcaps_array_vp3_rf nf=20 lf=20u bm=1 ns=4
array=1 m=5

Example 2: For VP4 MOM (Refer to Figure 1-1 (b))

x1 (plus minus nwell sub) momcaps_array_vp4_rf nf=21 lf=20u bm=1 ns=4
array=1 m=5

Example 3: For VP5 MOM (Refer to Figure 1-1 (c))

x1 (plus minus sub) momcaps_array_vp5_rfvcl nf=20 lf=20u bm=1 ns=4
array=1 m=5

For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo & ModelFileName.mdl.eldo into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "/ModelFileName.lib.eldo" CornerSection
, where CornerSection could be ff, tt, ss.
- Follow the example below to call the model

Example 1: For VP3 MOM (Refer to Figure 1-1 (a))

x1 plus minus nwell sub momcaps_array_vp3_rf nf=20 lf=20u bm=1 ns=4
array=1 m=5

Example 2: For VP4 MOM (Refer to Figure 1-1 (b))

x1 plus minus nwell sub momcaps_array_vp4_rf nf=21 lf=20u bm=1 ns=4
array=1 m=5

Example 3: For VP5 MOM (Refer to Figure 1-1 (c))

x1 plus minus sub momcaps_array_vp5_rfvcl nf=20 lf=20u bm=1 ns=4 array=1
m=5

ModelFileName for VP3 MOM: L110AE_MOMCAPS_ARRAY_VP3_RF_V022

ModelFileName for VP4 MOM: L110AE_MOMCAPS_ARRAY_VP4_RF_V022

ModelFileName for VP5 MOM: L110AE_MOMCAPS_ARRAY_VP5_RF_V022

As stated above that one scalable model is provided and this model is valid within the following geometry ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range: (before shrinkage)

For VP3 MOM:

Number of Finger (nf): 10 ~ 40 (Only even integer is allowed!)

Finger length (lf): 10 ~ 40 um(before shrinkage)

Bottom metal (bm): 1 ~ 4

Number of stack metal (ns): $3 \leq ns \leq (6 - bm + 1)$

Array size (array): 1 ~ 4

For VP4 MOM:

Number of Finger (nf): 11 ~ 41 (Only odd integer is allowed!)

Finger length (lf): 10 ~ 40 um(before shrinkage)

Bottom metal (bm): 1 ~ 4

Number of stack metal (ns): $3 \leq ns \leq (6 - bm + 1)$

Array size (array): 1 ~ 4

For VP5 MOM:

Number of Finger (nf): 10 ~ 40 (Only even integer is allowed!)

Finger length (lf): 10 ~ 40 um(before shrinkage)

Bottom metal (bm): 1 ~ 4

Shielding metal: bm-1

Number of stack metal (ns): $3 \leq ns \leq (6 - bm + 1)$

Array size (array): 1 ~ 4

Frequency range:

1 MHz ~ 0.8*Resonance(Max 16 GHz)

Capacitance range @ 1GHz:

For VP3 MOM:

28.6fF ~ 3.8pF

For VP4 MOM:

25.9fF ~ 3.65pF

For VP5 MOM:

28.6fF ~ 3.8pF

Temperature range:

-55 °C ~ +125 °C

When array ≥ 2 , all the ports need to be hooked to avoid the high resistance induced by the narrow metal lines used for port connection. The following example showed the correct port hookup for array=4 (P1:Minus, P2:Plus). The different port hookup will cause a quite different result with the model predicted.

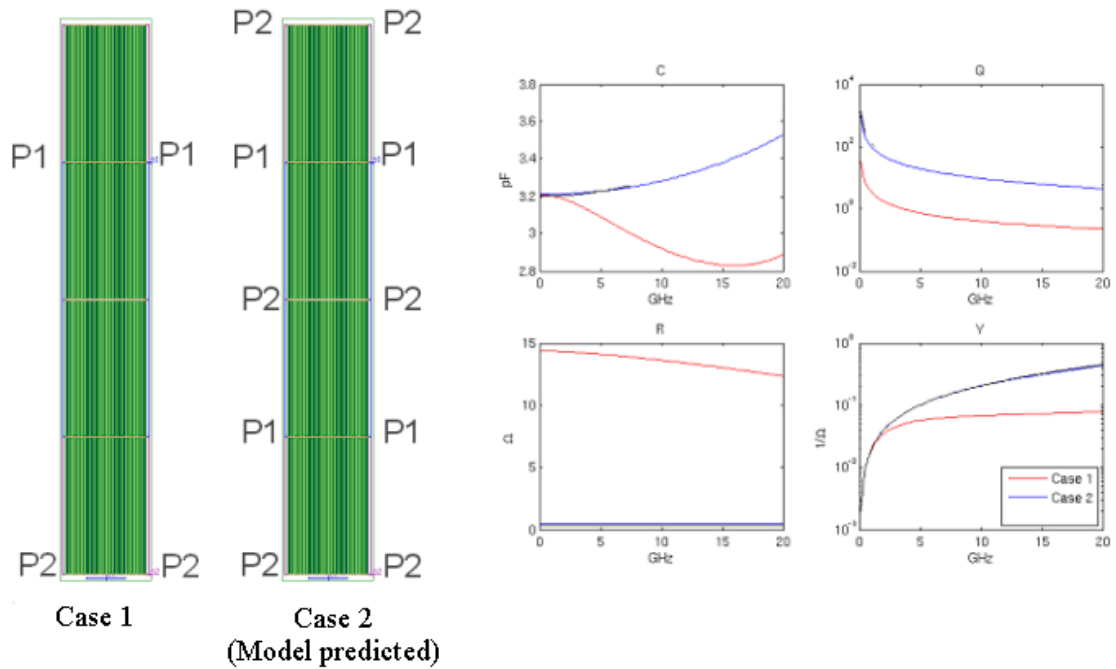


Figure 1-3 The ports' hook up of RF MOM

2 RFMOM Capacitors

2.1 Silicon Wafer for modeling

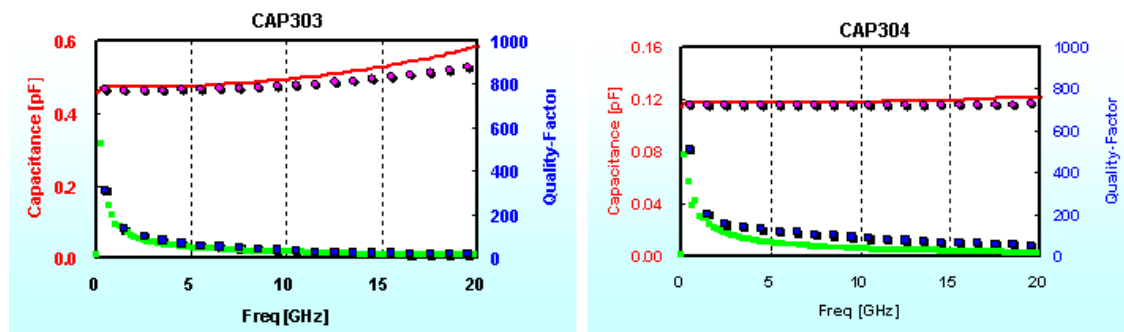
The information about the silicon wafer used for characterization and model parameter extraction is summarized below.

Table 2-1 Test wafer information

	MOMCAP
Mask ID	BSWFA
Lot ID	D4TK3
Wafer ID	#9
Metal option	1P7M

Table 2-2 Testkey list

Name	Length (um)	Number of fingers	Spacing per finger (um)	Width of finger (um)	Multiplier	Stacking
MOM303	40	20	0.2	0.2	1	M1-M6
MOM304	20	10	0.2	0.2	1	M1-M6
MOM305	20	40	0.2	0.2	1	M1-M6
MOM306	20	20	0.2	0.2	1	M1-M6
MOM307	20	20	0.2	0.2	1	M1-M5
MOM308	20	20	0.2	0.2	1	M1-M4
MOM503	40	20	0.2	0.2	1	M1-M6
MOM505	20	40	0.2	0.2	1	M1-M6



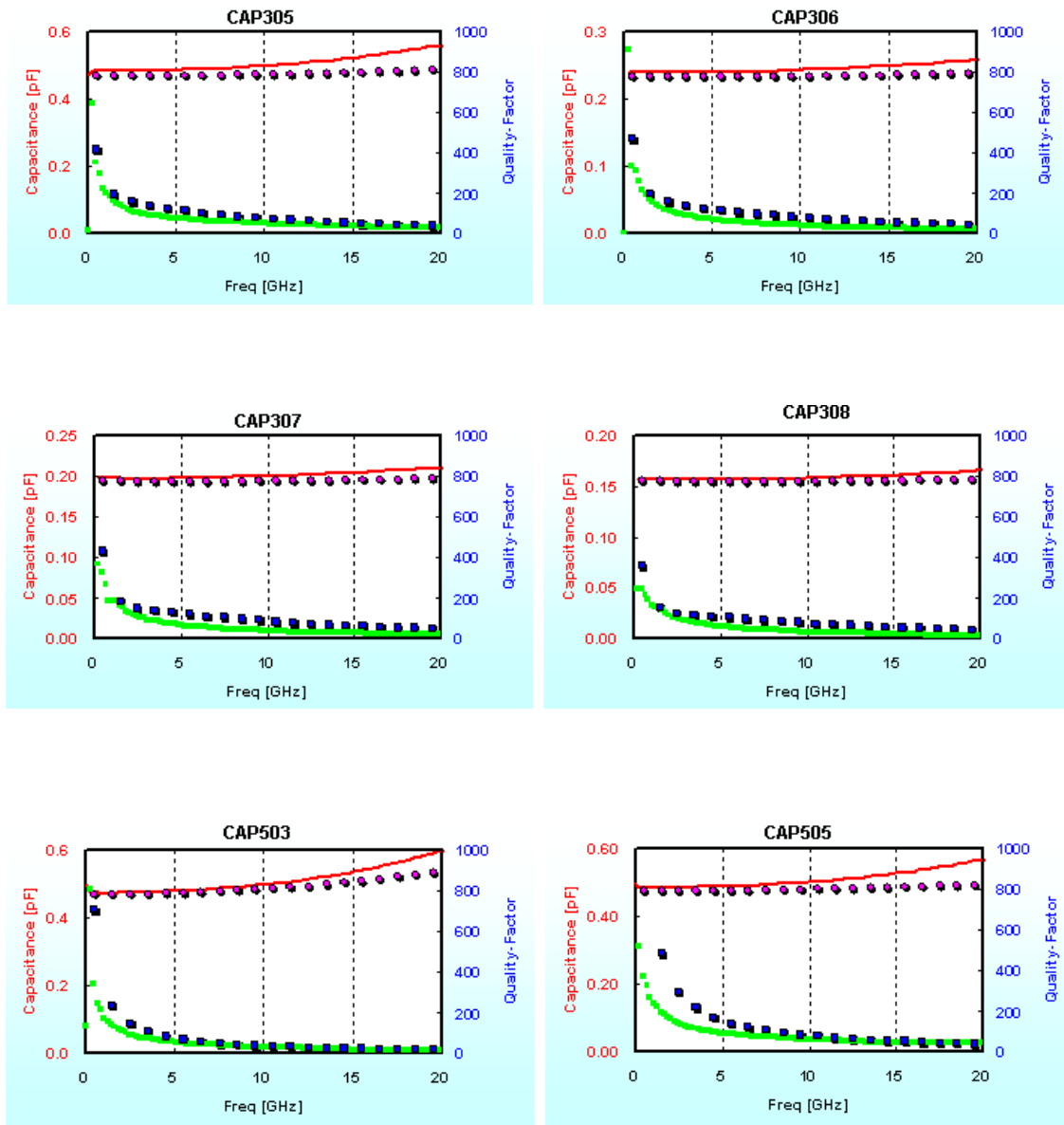


Figure 2-1 The EM calibration result

2.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized below.

$$C(f) = 1 / (2 \cdot \pi \cdot f \cdot \text{imag}(\frac{1}{y_{12}}))$$

$$Q(f) = \text{imag}(\frac{1}{y_{12}}) / \text{real}(\frac{1}{y_{12}})$$

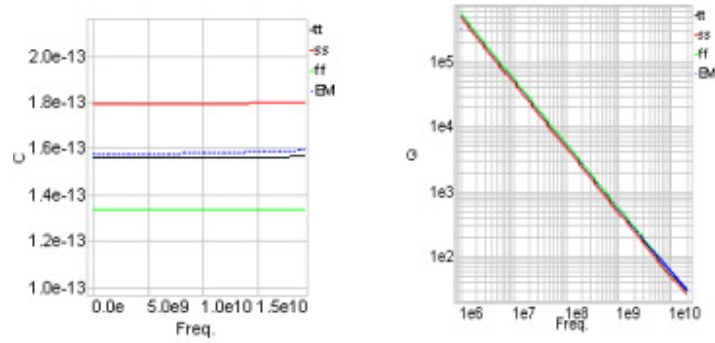
2.3 RFMOM Model

Model verification plots

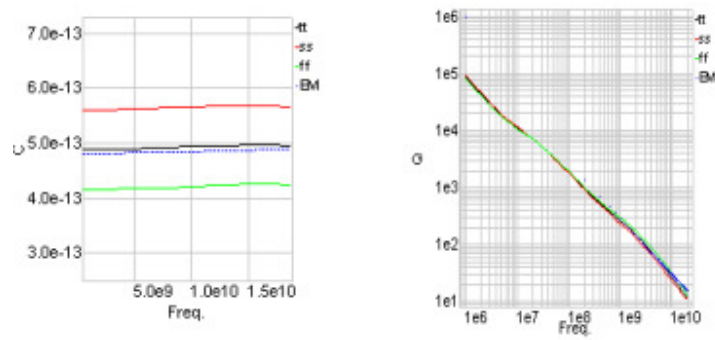
The simulation results (solid line) obtained from the extracted model were plotted together with the EM results (dotted line) for comparison.

C vs. Frequency and Q vs. Frequency plots

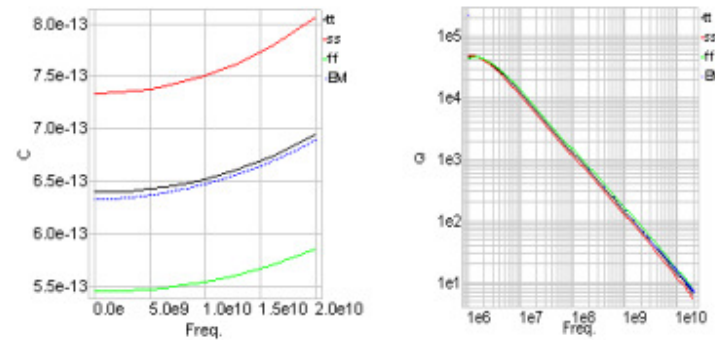
The following plots show the result of model and EM at 25 °C.



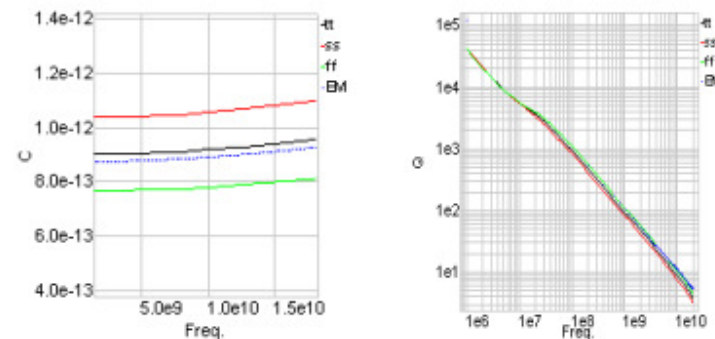
lf20u-nf10-ns4-bm2-array2



lf20u-nf20-ns4-bm3-array3

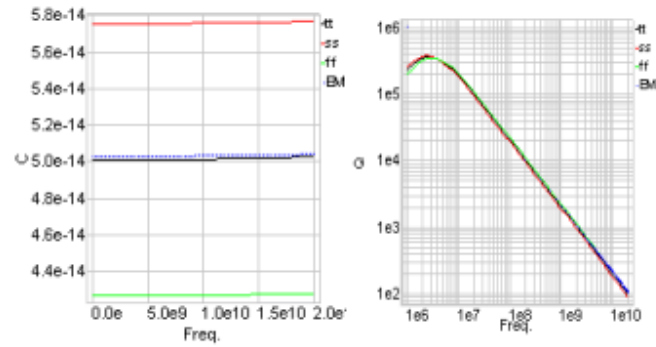


lf40u-nf20-ns4-bm2-array2

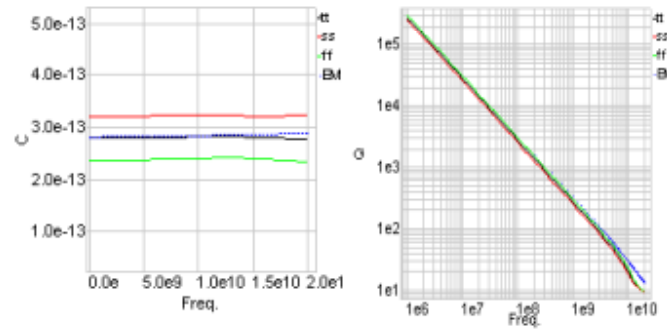


lf30u-nf30-ns5-bm1-array2

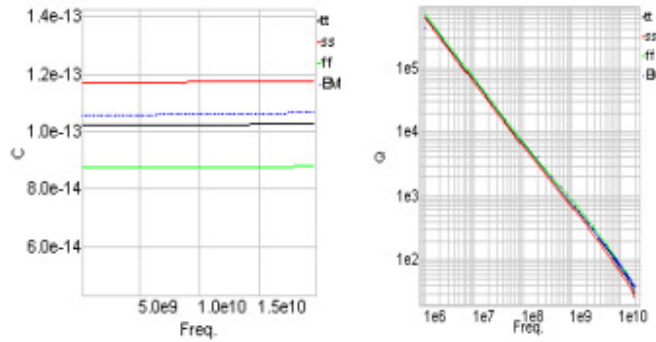
Figure 2-2 CQ plot of VP3



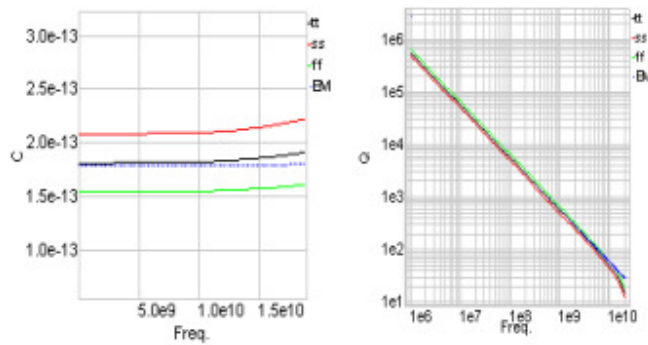
lf10u-nf21-ns3-bm1-array1



lf30u-nf21-ns5-bm2-array1

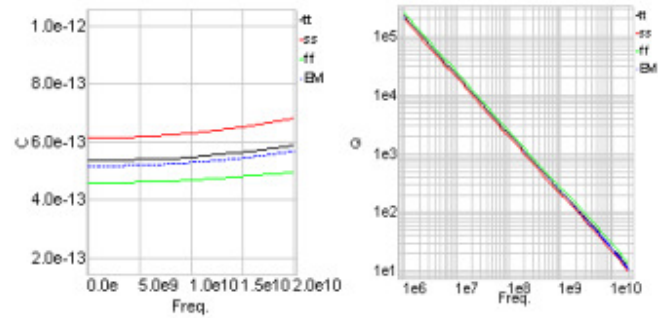


lf20u-nf11-ns5-bm1-array1

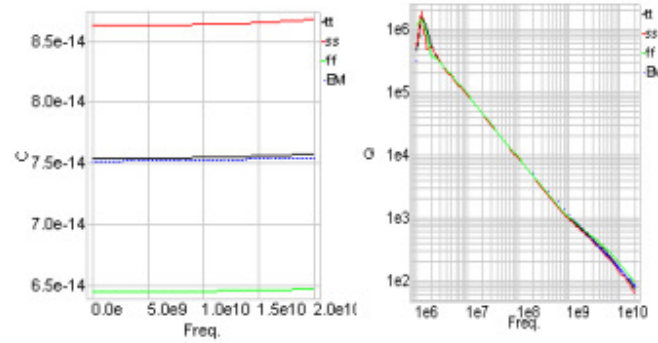


lf10u-nf31-ns6-bm1-array1

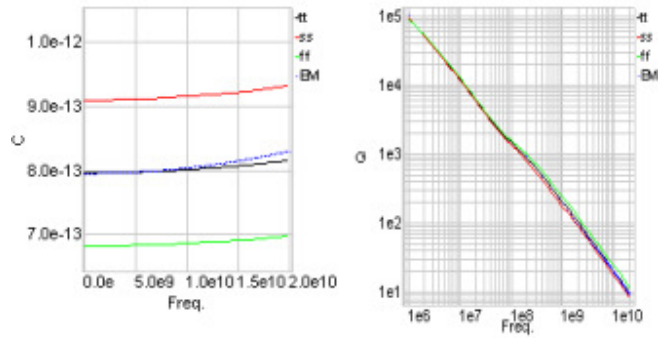
Figure 2-3 CQ plot of VP4



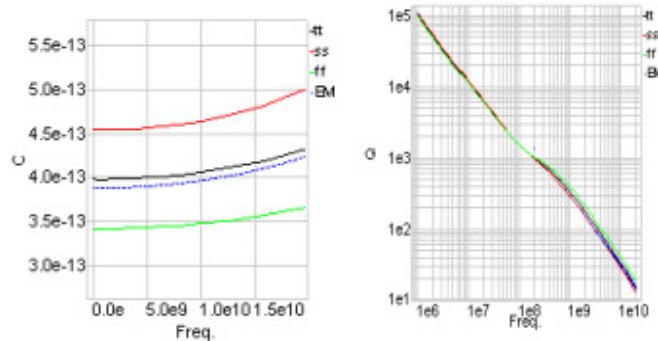
If30u-nf30-ns6-bm1-array1



If10u-nf10-ns4-bm1-array2



If30u-nf30-ns3-bm3-array3



If35u-nf40-ns3-bm1-array1

Figure 2-4 CQ plot of VP5

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	NA	NA

3.2 ELDO Warning Message

Table 3-2 Eldo warning message

NO.	Warning Message	Explanation
1.	NA	NA

3.3 SPECTRE Warning Message

Table 3-3 Spectre warning message

NO.	Warning Message	Explanation
1.	NA	NA